Branch Coverage Report

Project: Customer Churn Prediction (Team Hackforge)
Course: Software Engineering Mini Project, Deliverable Part 3
Date of test run: 30 October 2026
Generated by: `python tools/generate_test_reports.py` from a single run of the automated test suite

1. Method

Branch coverage was measured with coverage.py in branch mode (`--cov-branch`). Every decision point (`if`, `for`, `while`, `try`, conditional expressions) has two or more possible outcomes, and a branch is covered when the test suite takes that outcome at least once. This is a stricter measure than statement coverage. Test files and tooling are excluded.

2. Overall Result

Branches	Covered	Not taken	Partially covered decisions	Branch coverage
350	278	72	52	79.4%

3. Branch Coverage by Module

Module	Branches	Covered	Partial decisions	Branch coverage
`01_churn_prediction_xgboost.py`	10	9	1	90.0%
`02_paradigm_benchmark.py`	8	7	1	87.5%
`app.py`	122	88	18	72.1%
`classical_models.py`	0	0	0	n/a
`customer_presets.py`	2	2	0	100.0%
`dml_model.py`	18	11	5	61.1%
`prediction_pipeline.py`	176	149	25	84.7%
`qml_model.py`	2	2	0	100.0%
`training_common.py`	12	10	2	83.3%

4. Branches Not Taken

Each entry reads `source line -> destination line` (a negative destination means the function exits).

Module	Branches not taken
`01_churn_prediction_xgboost.py`	127->128
`02_paradigm_benchmark.py`	224->225

Module	Branches not taken
app.py	381->382, 502->503, 572->574, 597->600, 602->-578, 699->700, 724->725, 742->743, 760->761, 778->779, 885->886, 962->963, 1079->1080, 1124->1125, 1165->1166, 1172->1173, 1172->1175, 1183->1184, 1183->1191, 1192->1193, 1192->1199, 1199->-1, 1199->1200, 1215->-1, 1215->1216, 1216->1217, 1216->1227, 1228->-1, 1228->1229, 1284->1285, 1284->1286, 1362->1365, 1381->1384, 1404->-1
classical_models.py	none
customer_presets.py	none
dml_model.py	67->69, 76->-64, 202->205, 206->197, 206->207, 241->243, 248->-245
prediction_pipeline.py	220->226, 221->222, 234->235, 340->341, 355->361, 393->397, 398->399, 410->409, 457->458, 497->502, 528->531, 613->614, 618->617, 620->621, 653->657, 683->685, 735->744, 756->765, 773->782, 796->805, 838->839, 840->844, 851->853, 856->859, 873->875, 875->876, 875->877
qml_model.py	none
training_common.py	81->83, 245->264

The branches not taken are mainly defensive error paths (for example a corrupt pickle file) and optional user interface paths. The core decision logic in input validation, preprocessing, prediction and training is exercised with both valid and invalid input.